The Shooting Solver

Independent verification and refinement of the temporal stability eigenvalue by initial-value integration in `pyMack`

`pyMack` technical documentation

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Contents

1 Scope

The companion numerical-methods document solves the temporal stability problem as a boundary-value computation: one large matrix pencil, assembled by Chebyshev collocation, handed to a dense eigensolver that returns all of its eigenvalues at once. The present document poses the identical question (for a given real wavenumber α, at what complex phase speed c does a non-trivial disturbance satisfy decay at the freestream and the wall conditions?) as an *initial-value* computation [2]: the disturbance equations are recast as a small first-order system, integrated from the freestream to the wall, and the eigenvalue is located by driving a 3×3 wall matrix singular. No global matrix is ever formed; the cost per evaluation is small, but a good initial guess is required. This is the classical shooting procedure of Mack [1], in the formulation of Özgen & Kırcalı [3]. Within `pyMack` its role is verification and refinement: given a mode located by the spectral solver, shooting confirms it through entirely independent numerics and sharpens it to tight tolerance.

Section 2 recasts the disturbance equations as the six-component first-order system; Section 3 states the wall conditions and constructs the decaying freestream basis; Section 4 marches that basis to the wall; Section 5 converts the wall conditions into the singularity condition on the wall matrix; Section 6 locates the eigenvalue; and Section 7 places the method against the spectral solver. Figure 1 shows the complete pipeline.

Notation. Overbars denote mean (base-flow) quantities; hats denote complex disturbance amplitudes. α is the real streamwise wavenumber, $c = c_r + \mathrm{i}c_i$ the complex phase speed, M_e, Re, Pr, γ the edge Mach number, Reynolds number, Prandtl number, and specific-heat ratio. Y is the six-component shooting state (a 6×3 matrix once three solutions are tracked), $A(y)$ its pointwise 6×6 coefficient matrix, $M(c)$ the 3×3 wall matrix, and $\sigma_{\min}(c)$ its smallest singular value.

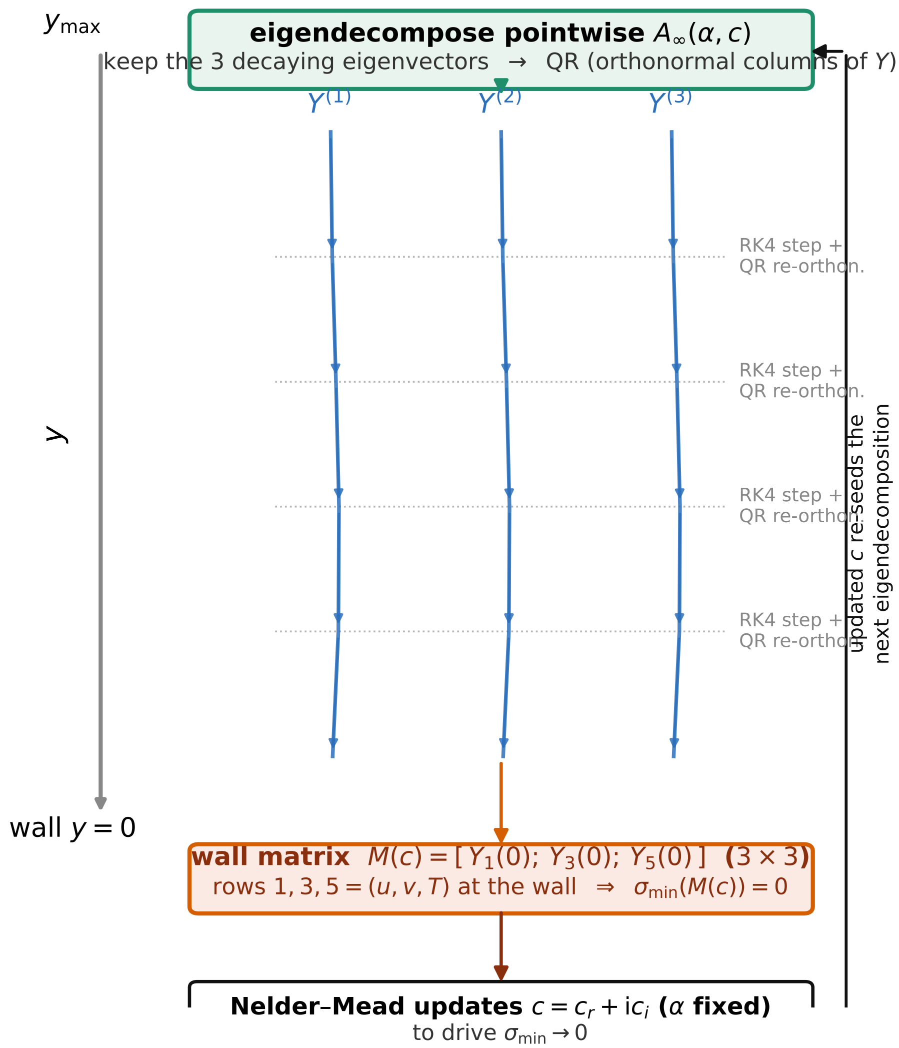

Figure 1: The shooting pipeline, top ($y_{\max}$) to bottom (wall, $y = 0$). At $y_{\max}$ the constant-coefficient matrix A_∞ is eigen-decomposed and its three decaying eigenvectors are QR-orthonormalised into the starting 6×3 basis (Section 3); the basis is RK4-marched to the wall with re-orthonormalisation after every step (Section 4); at the wall the three arriving columns form the 3×3 matrix $M(c)$ (Section 5); and an outer Nelder–Mead loop (Section 6) updates the trial c to drive $\sigma_{\min}(c) \to 0$, feeding back into the eigen-decomposition at $y_{\max}$.

2 The first-order state and its coefficient matrix

Given a real wavenumber α and a trial phase speed c, the four disturbance equations are cast as a single first-order ODE in a six-component complex state. Following Özgen & Kırcalı [3, Eqs. 22–28], specialised to two dimensions ($\beta = 0$, no spanwise velocity), the state is

$$Y(y) = \big(X_1, X_2, X_3, X_4, X_5, X_6\big)^{\mathsf{T}} \in \mathbb{C}^6, \tag{1}$$

in which X_1, X_3, X_5 are the primary flow amplitudes, X_2, X_6 the wall-normal derivatives of X_1, X_5, and X_4 the pressure amplitude. Concretely, $X_1 = \alpha\hat{u}$ is the wavenumber-scaled streamwise-velocity amplitude (the k-direction combination $\alpha\hat{u} + \beta\hat{w}$ of [3] evaluated at $\beta = 0$; the constant factor α is immaterial in a homogeneous eigenvalue problem), $X_3 = \hat{v}$ the wall-normal velocity, $X_5 = \hat{T}$ the temperature (with $X_2 = \mathrm{D}X_1,\ X_6 = \mathrm{D}X_5$), and $X_4 = \hat{p}$ the same $\rho_e U_e^2$-scaled pressure as in the spectral solver, coupled through the continuity and y-momentum rows rather than as a plain derivative. The state thus stacks $(\alpha u,\ \alpha u',\ v,\ p,\ T,\ T')$, and the system closes as one first-order ODE,

$$\frac{\mathrm{d}Y}{\mathrm{d}y} = A\big(y;\ \alpha, c, Re, M_e\big)\, Y, \qquad A \in \mathbb{C}^{6\times6}, \tag{2}$$

where A is assembled *pointwise*, one 6×6 matrix at each y, from the local mean-flow data $\overline{U}, \overline{U}', \overline{U}''$, $\overline{T}, \overline{T}'$, $\overline{\mu}$ with $\mathrm{d}\mu/\mathrm{d}T$, $\mathrm{d}^2\mu/\mathrm{d}T^2$, and $\overline{\kappa}$ with its temperature derivatives. Rows 1 and 5 of A carry the trivial derivative definitions $\mathrm{D}X_1 = X_2$ and $\mathrm{D}X_5 = X_6$ (single entries $A_{12} = A_{56} = 1$); the physics resides in rows $2, 4, 6$ (the two momentum balances and the energy balance) and in row 3, continuity with the equation of state.

The pointwise structure mirrors the block operator of the spectral solver term for term: the convective factor $\alpha(\overline{U} - c)$, the viscous prefactor $\propto Re/(\overline{\mu}\overline{T})$, the transport-derivative couplings $\frac{\mathrm{d}\mu}{\mathrm{d}T}\overline{U}'$, and the conduction group in $\overline{\kappa}$ all reappear, now as scalar entries of a 6×6 matrix at one height rather than as $n \times n$ blocks. The gas defaults ($\gamma = 1.4$) and the length scaling (L^*) match the spectral solver, so both routes see the identical mean flow.

Remark: a correction to the printed reference. The 6×6 matrix is evaluated from the general Appendix-A formulation of Mack [1] at $\beta = 0$ with $\lambda/\mu = 0$ (the same Stokes closure as the temporal spectral solver), which at those settings is algebraically identical to the printed Eqs. 22–28 of [3], term for term, with one correction: the X_1 coefficient of the printed Eq. (24) reads $+\mathrm{i}\big(2\mu\frac{\mathrm{d}\mu}{\mathrm{d}T}T' + \frac{4}{3}\frac{T'}{T}\big)$, but re-deriving the pressure row from Eqs. (13) and (23) of the same reference gives $-\mathrm{i}\big(2\frac{1}{\mu}\frac{\mathrm{d}\mu}{\mathrm{d}T}T' + \frac{4}{3}\frac{T'}{T}\big)$ (sign and μ-placement). With the corrected term the two solvers agree in practice: at the reference parameters of the companion document's worked example ($M_e = 6$, $R = 5500$, $\alpha_L = 0.174$), $\sigma_{\min}(c)$ falls to 4×10^{-6} at the spectral eigenvalue $c = 0.930 + 0.0200\mathrm{i}$ and rises by two orders of magnitude within $|\Delta c| \sim 2\times10^{-3}$ of it.

3 Boundary conditions: the wall and the decaying freestream basis

Six conditions pin down a non-trivial disturbance: three homogeneous conditions at the wall, and three at the freestream selecting only the decaying solutions.

Wall conditions. At $y = 0$ the three primary amplitudes vanish,

$$X_1(0) = 0 \text{ (no slip)}, \qquad X_3(0) = 0 \text{ (no penetration)}, \qquad X_5(0) = 0 \text{ (isothermal wall)}, \tag{3}$$

the pointwise counterparts of the wall identity rows of the spectral assembly.

Freestream: the decaying basis. As $y \to y_{\max}$ the mean flow becomes uniform and the coefficient matrix approaches a constant, $A_\infty = A(y_{\max})$. A constant-coefficient system $\mathrm{D}Y = A_\infty Y$ has solutions $Y \propto e^{\lambda y} v$ built from the eigenpairs (λ, v) of A_∞; the six eigenvalues split into three with $\operatorname{Re}\lambda > 0$, which grow outward and are unphysical, and three with $\operatorname{Re}\lambda < 0$, which decay outward as a bounded mode requires. Only the three decaying eigenvectors are retained,

$$A_\infty v_k = \lambda_k v_k, \qquad \operatorname{Re}(\lambda_k) < 0, \quad k = 1, 2, 3, \tag{4}$$

assembled into a 6×3 matrix and orthonormalised by a QR factorisation into the starting basis $Y(y_{\max})$. Enforcing decay in this way is the shooting analogue of the spectral solver's freestream identity rows. If a trial c yields fewer than three strictly decaying eigenvalues, the three most decaying (smallest $\operatorname{Re}\lambda$) are taken so the march remains well defined; near the physical eigenvalue exactly three decay.

4 Integrating the bounded basis to the wall

The three admissible freestream solutions must be propagated to the wall so their arrival values can be tested against (3). Because three independent solutions are tracked simultaneously, the state is the 6×3 matrix Y whose columns are the three solutions. It is marched from $y_{\max}$ down to $y = 0$ on a fixed grid of about 800 steps with the classical fourth-order Runge–Kutta scheme; over one step of size h (here $h < 0$),

$$\begin{aligned} k_1 &= A(y_0)\, Y, & k_2 &= A(y_m)\big(Y + \tfrac{1}{2} h\, k_1\big), \\ k_3 &= A(y_m)\big(Y + \tfrac{1}{2} h\, k_2\big), & k_4 &= A(y_1)\big(Y + h\, k_3\big), \end{aligned} \qquad Y \leftarrow Y + \frac{h}{6}\big(k_1 + 2k_2 + 2k_3 + k_4\big), \tag{5}$$

with $y_m = \frac{1}{2}(y_0 + y_1)$ the step midpoint and $A(\cdot)$ the pointwise matrix of Section 2.

Re-orthonormalisation. The three solutions decay at very different rates, so after a short march the rapidly decaying columns are numerically swamped by the slowest one and the basis collapses toward a single direction, the parasitic-error pathology classical in shooting for stiff stability problems [4, 1]. To suppress it, after every RK4 step the columns are re-orthonormalised by a QR factorisation,

$$Y \longleftarrow Q, \qquad \text{where} \quad Y = QR \;\; (Q \in \mathbb{C}^{6\times 3} \text{ orthonormal}), \tag{6}$$

retaining only the orthonormal factor; the discarded triangular factor merely records scaling. The operation preserves the *subspace* spanned by the columns while keeping them well conditioned, and it is that subspace, not the individual column magnitudes, on which the eigenvalue condition below depends.

5 The eigenvalue condition and the wall matrix

At the wall the integrated basis is a 6×3 matrix $Y(0)$, and any admissible disturbance is a combination $Y(0)\,a$ of its columns, $a \in \mathbb{C}^3$. The wall conditions (3) require rows $1, 3, 5$ of that combination to vanish; collecting those rows gives the 3×3 *wall matrix*

$$M(c) = \begin{bmatrix} Y_1(0) \\ Y_3(0) \\ Y_5(0) \end{bmatrix} \in \mathbb{C}^{3\times 3}, \qquad \text{wall conditions:} \quad M(c)\,a = 0. \tag{7}$$

A non-trivial $a \neq 0$ exists only when $M(c)$ is singular. Rather than testing a determinant, which is poorly scaled after the repeated re-orthonormalisation, the smallest singular value of $M(c)$ is monitored,

$$\sigma_{\min}(c) = \min \operatorname{svd}\big(M(c)\big), \qquad \boxed{\sigma_{\min}(c) \to 0 \iff c \text{ is an eigenvalue}}, \tag{8}$$

a non-negative, well-scaled measure of distance to singularity: it vanishes exactly when the three decaying solutions can be combined to satisfy all three wall conditions, that is, exactly when a genuine boundary-layer mode exists at that c.

6 The eigenvalue search

It remains to locate the complex $c = c_r + \mathrm{i}c_i$ at which $\sigma_{\min}(c) = 0$, starting from a physics-based guess; this single c is the temporal eigenvalue, identical to the one the spectral solver returns. Since c is complex, driving $\sigma_{\min} \to 0$ is a two-dimensional real minimisation over (c_r, c_i). `pyMack` applies the Nelder–Mead simplex method [5] to the objective $\sigma_{\min}$, from an initial guess such as the second-mode estimate $c_r \approx 1 - 1/M_e$. The search is confined to the physical strip

$$c_r \in (0,\, 1.2), \qquad c_i \in (-0.2,\, 0.2), \tag{9}$$

enforced by a large penalty (10^6) outside the box, and converges to the tolerances

$$|\Delta c| < 10^{-7}, \qquad \Delta\sigma_{\min} < 10^{-8}, \tag{10}$$

within at most ∼120 iterations. Each objective evaluation runs the entire pipeline of Sections 3–5 (Figure 1): build the freestream decay basis, integrate with re-orthonormalisation to the wall, form $M(c)$, take $\sigma_{\min}$. The converged minimiser is the eigenvalue, and the growth rate follows exactly as in the spectral solver, $\omega_i = \alpha c_i$.

7 Spectral versus shooting

The two solvers answer the identical question (for a given real α, at what complex c does a non-trivial disturbance decay at the freestream and satisfy the wall conditions?) and return the same physical phase speed, but with complementary characteristics, mirroring the classical boundary-value/initial-value trade-off [2]:

	Spectral (companion document)	**Shooting (this document)**
core object	one $4n \times 4n$ pair (A, B)	one 6×6 matrix $A(y)$, evaluated pointwise
what it returns	*all* $4n$ eigenvalues at once	*one* eigenvalue near a guess
freestream decay	identity rows on the edge nodes	the 3 decaying eigenvectors of $A(y_{\max})$
wall conditions	identity row replacement	$M(c)$ singular, $\sigma_{\min}(c) \to 0$
how the answer is found	dense QZ / shift–invert	RK4 march + QR, then Nelder–Mead on $\sigma_{\min}$
initial guess	not required	required (physics-based)
memory / cost	large dense matrices, $O((4n)^3)$	tiny system, cheap per evaluation
best used for	surveying the spectrum; sweeping neutral curves and N-factors	pinpointing one mode to tight tolerance; independent verification

In practice the two are used together: the spectral solver locates a mode without prior knowledge of the spectrum, and the shooting solver refines it and confirms it through entirely independent numerics. Agreement between the two on c is strong evidence that the eigenvalue is physics rather than a discretisation artifact.

References

[1] L. M. Mack, "Boundary-layer linear stability theory," AGARD Report No. 709, *Special Course on Stability and Transition of Laminar Flow*, 1984.

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[4] S. D. Conte, "The numerical solution of linear boundary value problems," *SIAM Review*, Vol. 8, No. 3, 1966, pp. 309–326.

[5] J. A. Nelder and R. Mead, "A simplex method for function minimization," *The Computer Journal*, Vol. 7, No. 4, 1965, pp. 308–313.